Raynham Estate OS

Product Requirements Document v2.0 · Build Specification · July 2026

North Star A living spatial picture of the estate. Tom sees the estate from above; staff update it from where they stand.

Status: Build-ready product direction. This document supersedes the previous North Star PRD where requirements conflict. The latest desktop and mobile mock-ups define visual composition; this PRD defines product behaviour, architecture, data, safety logic, delivery phases and acceptance criteria.

Primary audience: AI development agents, product designers, engineers, QA, estate operations stakeholders.

1. Product in one page

Raynham Estate OS is a spatial operations platform for a working rural estate. The estate itself is the primary interface: people, places, assets, work, notices and risk are attached to physical location and viewed in context.

Three jobs

Job	User outcome
Keep people safe	A worker's phone becomes the work-time safety beacon. Lone-work sessions, check-ins, recent location and emergency escalation are visible to management.
Report anything	A worker can report from the place they are standing, or scan an exact QR-tagged asset, add a photo/category and submit in under 30 seconds.
See the estate	Tom opens one whole-estate map and sees current people, lone workers, issues, assets, notices, risks and what needs attention.

Product principle

The estate is the application Reports, Safety and Assets are not isolated destinations. They are layers and states on one spatial model. Lists and panels exist to act on the map, not replace it.

2. Goals, non-goals and success

Goals

- Replace the practical role of the dedicated lone-worker fob with a phone-centred workflow people are more likely to carry.
- Reduce the time between spotting an issue and the right person seeing, assigning and resolving it.
- Give Tom a calm, whole-estate operating picture rather than a conventional dashboard.
- Create a persistent spatial history for locations and assets: what happened here, what is open, what recurs, what was fixed.
- Work in rural conditions: bright light, gloves, poor GSM, interrupted connectivity and older devices/users.

Non-goals for v2.0

- Accounting, payroll, stock control, farm ERP or property-management ERP.
- Dense BI dashboards, chart-heavy reporting or enterprise workflow configuration.
- Continuous off-duty employee tracking.
- Claiming that a PWA alone provides certified fall detection. Device-sensor/fall-detection features require a later technical validation/native capability track.
- Guest-facing hospitality product beyond simple anchor information unless separately approved.

Success measures

Metric	Target / intent
Report completion	Typical field report in <30 seconds with no typing required.
Location confidence	Every operational item carries source + timestamp + location confidence/fallback state.
Management comprehension	Tom can answer "Is the estate OK, what needs attention, and where?" from the opening screen.
Safety clarity	Lone-worker state and stale location are explicit; red is reserved for emergency.
Offline resilience	Reports queue locally and sync automatically when connectivity returns.

3. Users and permission model

Role	Primary experience	Permissions
Owner — Tom	Whole-estate spatial view	All workspaces, people, work, safety, notices, history, configuration and export.
Workspace Manager	Map filtered to Farm & Land, Property & Grounds or Hospitality	Triage, assign, publish notices, manage team/anchors in workspace, see permitted worker safety state.
Field Staff	Mobile "estate from where I stand"	Working presence, report, scan, lone-work session, check-in, help.
Contractor	Link-based task experience	View assigned job/location; update/resolve without full account where appropriate.
Tenant / reporter	QR/mobile report flow	Read anchor info, report, request help; identity optional unless safety session requires it.

Privacy rule: work-location sharing must have a visible state and a clear stop boundary. Off-duty users are shown as off duty, not tracked on the estate map. Exact retention, lawful basis, staff policy and emergency procedures must be agreed before production rollout.

4. Core spatial model

The central object is not the report; it is the physical world. Every operational record must resolve to a place, asset, person or combination of them.

Object	Definition	Key relationships
Estate	Raynham as the top-level spatial domain.	Contains workspaces and zones.
Workspace	Operational lens, not a silo.	Farm & Land; Property & Grounds; Hospitality.
Zone	Named physical area.	North Field, Home Farm, North Woods, Cottage area.
Place / Anchor	Fixed physical location; anchor may have QR.	Zone, GPS, What3Words, notices, history.
Asset	Physical thing that may move or be fixed.	Anchor optional; current/last-known location; history.
Person	Worker identity and work state.	Device, workspace, presence, lone-work sessions.
Observation / Report	What someone saw at a place/time.	Reporter, GPS, photo, category, asset/anchor optional.
Work Item	Action created from an observation or directly.	Assignment, status, due/priority, resolution evidence.
Notice	Context someone should see at a place/asset.	Expiry mandatory; visible on map and QR scan.
Safety Event	Emergency or overdue safety state.	Person, session, last-known location, escalation trail.

Four location concepts — never merge them

1. Worker location: recent position reported by the worker's phone. It moves and must show age/staleness.
2. Incident location: fixed position where an observation/report occurred.
3. Asset location: registered/current/last-known position of an asset; may be inferred but must label the source.
4. Anchor location: fixed surveyed/registered position of a QR/location anchor.

Location truth rule Never display a stale or inferred location as "live". Store timestamp, source and confidence. UI labels must use "updated X ago", "last seen", or "location unavailable" when appropriate.

5. Information architecture

Desktop — Tom / managers

- Default route: whole-estate map for Tom; workspace-filtered map for managers.
- Header: Raynham, workspace lens, search, profile/admin.
- Floating map controls: Layers + time range.
- Floating operational panels: Estate Status, Estate Brief, Needs Attention.
- Context side inspector appears only when a person/place/asset/issue is selected.
- "Work", "Safety", "Notices", registry/admin can exist as secondary views/drawers, but map context remains the system spine.

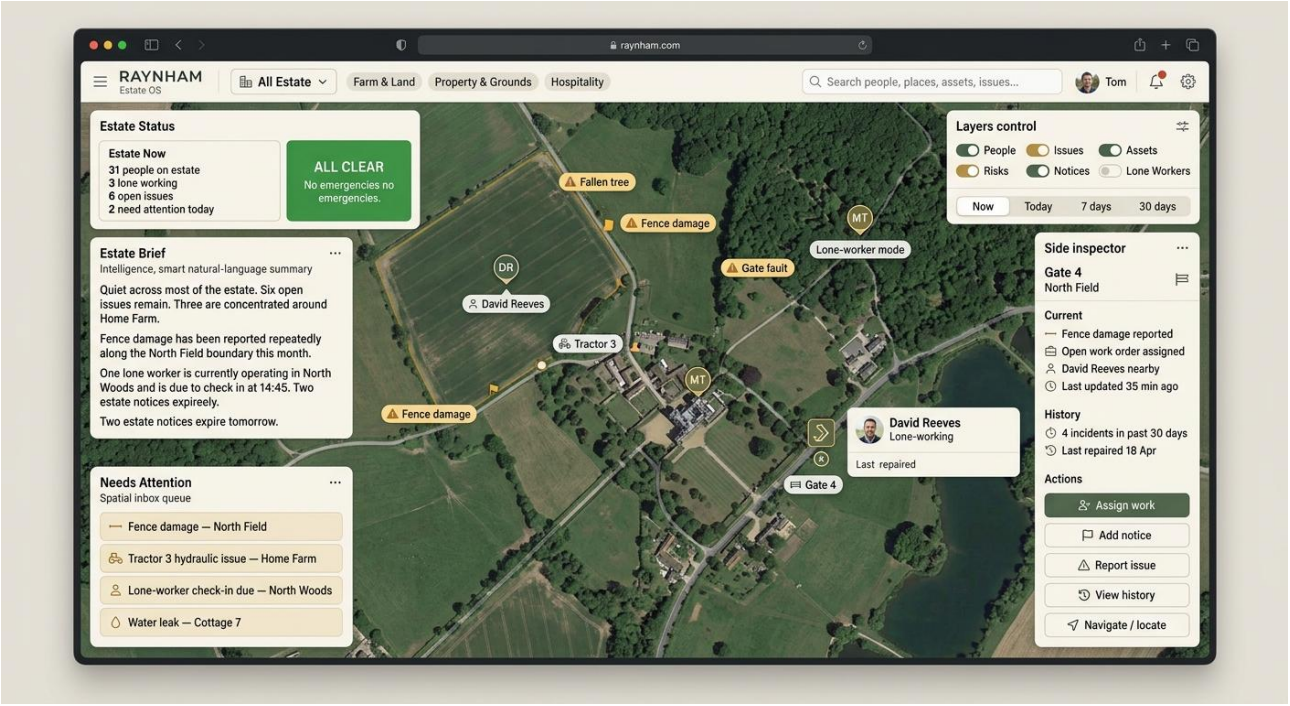
Mobile — staff / lone workers

- Home answers: where am I, am I connected, what is my work/safety state, what is nearby, what can I do here?

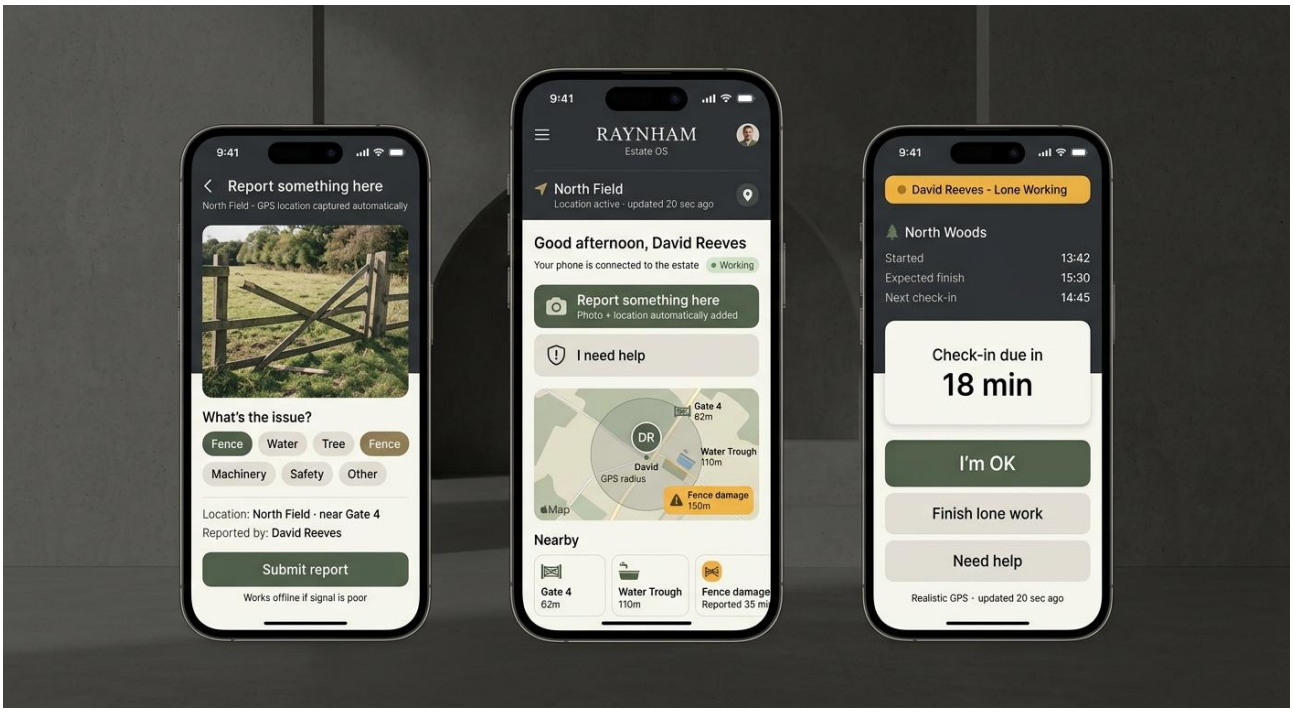
- Primary actions: Report something here; I need help.
- Compact map shows worker marker, GPS accuracy, nearby anchors/assets/issues.
- Lone Work is a distinct state/mode with expected finish, next check-in, I'm OK, finish and help actions.
- QR scan gives exact object identity; GPS gives spatial context. QR remains optional for general reporting.

6. Visual north star

The two approved concept mock-ups below are the visual reference for v2.0. They define composition and feel; this PRD defines behaviour and edge cases.



Desktop — whole-estate spatial operating picture



Mobile — worker presence, reporting and lone-worker mode

7. Desktop experience — Tom's spatial overview

7.1 Opening state

- Map occupies the majority of the viewport; satellite/hybrid imagery is the hero.
- Estate Status shows people on estate, lone workers, open issues, and "needs attention" count.
- ALL CLEAR means no active emergency and no critical/overdue safety escalation. It does not mean no maintenance work exists.
- Estate Brief is a concise natural-language operational summary with each statement linkable to its relevant map context.
- Needs Attention is a spatial inbox; selecting an item flies/zooms to its location and opens context.

7.2 Layers

Layers are independently composable. Do not implement mutually exclusive "Assets OR Reports OR Safety" tabs.

Layer	Examples	Default
People	Working staff, managers, contractors	On
Lone Workers	Active lone-work sessions	On
Issues / Work	Open observations/work items	On
Assets / Anchors	Tractors, gates, pumps, buildings, QR anchors	Key only
Notices	Active context notices	Off/summary
Risks	Active hazards/restricted areas	On when active

7.3 Zoom behaviour

Zoom	Render
Estate	Zone summaries/clusters; no pin soup.
Zone	People, open issues, major assets/anchors, risk areas.
Local	Precise anchors/assets/issues, tracks/buildings, detailed labels.

7.4 Side inspector

Clicking any mapped object opens a contextual panel while preserving the map. A Place/Asset inspector must support Current, History and Actions. Person inspector must show work/safety state, last location timestamp and permitted contact actions. Report/Work inspector must show lifecycle and related place/asset.

7.5 Search / command

- Search people, places, assets and issue text.
- Results fly to map context rather than opening disconnected records.
- Later phase: constrained natural-language queries such as "show unresolved maintenance near Home Farm"; do not block MVP on this.

8. Mobile experience — worker phone as beacon + field interface

8.1 Working state

When an identified worker starts work, the phone becomes their work-time digital presence. The system records location updates subject to browser/OS capability and declared tracking policy.

- Show zone/name derived from GPS: e.g. North Field.
- Show "Location active · updated 20 sec ago" or an honest fallback/stale state.
- Show worker identity and Working status.
- Primary: Report something here. Secondary: I need help.
- Nearby map/context displays only useful nearby objects/issues, with distance.

8.2 Report something here

5. Open Report Something Here.
6. Capture or select a photo. If camera denied, file picker is mandatory fallback.
7. Choose one category; optional note/voice input.
8. Submit. Auto-attach worker identity if known, GPS, timestamp, zone, What3Words if service available, nearest anchor/asset candidates and current lone-work session ID if applicable.
9. If offline, save to durable local outbox and immediately show confirmation with queued state; sync on reconnect.

8.3 QR scan

QR is for precision and physical identity. A scan resolves an Anchor/Asset and shows active notices before actions. From that object the worker can read info, report issue, or get help. Anonymous reporting remains supported; identity is required only for worker-presence/lone-work functionality.

8.4 Lone-worker mode

10. Worker starts Lone Work and confirms task/type plus expected finish time.
11. System creates an active safety session and starts configured check-in cadence.
12. UI shows location state, session start, expected finish, next check-in, I'm OK, Finish lone work, Need help.
13. Check-in resets the due timer and timestamps the event.
14. If a check-in/finish deadline is missed, enter Attention state first, then escalate according to configured policy.
15. Emergency action creates a Safety Event immediately with person, current/last-known location, time and escalation recipients.

Important technical constraint A web/PWA experience cannot be assumed to deliver reliable continuous background GPS or certified fall detection on every iOS/Android state. Phase 2 must validate background-location behaviour in the target devices. If continuous beaconing or device-sensor detection is mandatory, introduce a native wrapper/app capability instead of pretending the PWA can guarantee it.

9. Work lifecycle — separate reporting from doing

Field staff report observations; managers manage work. This avoids turning the field flow into a ticketing form.

State	Meaning	Owner
Observation received	A fact/problem was reported.	System/manager triage
Needs triage	Routing/duplicate/location needs confirmation.	Workspace manager
Assigned	A work item has an owner.	Staff/contractor

In progress	Work has started.	Assignee
Waiting	Blocked by part/person/weather/access.	Assignee/manager
Resolved	Assignee says work is complete.	Assignee
Verified / Closed	Optional manager verification; remains in spatial history.	Manager

A report may create a Work Item automatically based on routing rules, but the Observation record must remain immutable as the original evidence. Repeat-location detection should surface patterns without merging away original observations.

10. Safety model and states

State	Colour	Meaning
Working / Safe	Green	Normal active presence; no safety escalation.
Lone Working	Amber	Enhanced safety monitoring is active; not an emergency.
Attention	Amber	Check-in due/overdue within configurable grace window or location stale.
Emergency	Red	Explicit help request or escalation threshold reached. Red is reserved for this.
Off duty	Neutral	No work location sharing.

Minimum escalation configuration

- Check-in cadence by activity/risk type.
- Grace period before manager attention.
- Escalation recipients/on-call rota.
- Escalation channels available in implementation (push/SMS/email/phone workflow).
- Acknowledgement and resolution timestamps.
- Fallback when location permission/signal is unavailable: last-known location + age + contact details.

Before production, Raynham must define the real-world emergency SOP: who receives alerts, how quickly they act, when external emergency services are contacted, and what happens if the platform is unavailable. Product UI must not imply automatic emergency-service dispatch unless that service genuinely exists.

11. Data contracts — minimum build model

Use UUIDs internally; human references (e.g. RN-0427) are display identifiers. All timestamps ISO 8601 UTC, rendered in estate local time.

Entity	Required fields
Person	id, name, mobile, role, workspaceIds, status, deviceBinding, createdAt
Presence	personId, state, startedAt, endedAt?, locationPermission, lastLocationId?
LocationSample	id, personId?, lat, lng, w3w?, accuracyM?, source, capturedAt, receivedAt

Zone	id, workspaceId, name, geometry/bounds, centroid
Anchor	id, zoneId, name, lat, lng, w3w?, qrToken, assetId?, status
Asset	id, zoneId?, name, type, fixedOrMobile, registeredLocation?, lastKnownLocation?, anchorIds[]
Observation	id/ref, category, note?, photoIds[], reporterId?, lat/lng, zoneId, anchorId?, assetId?, createdAt
WorkItem	id, observationId?, title, workspaceId, assignee, state, priority, dueAt?, timestamps
Notice	id, scopeType/scopeId, body, severity, startsAt, expiresAt, authorId
LoneWorkSession	id, personId, activity, startedAt, expectedEndAt, nextCheckInAt, state, lastCheckInAt?
SafetyEvent	id, sessionId?, personId, type, triggeredAt, locationId?, state, acknowledgedAt?, resolvedAt?

API boundary

- UI must use a small service/API layer; no component writes directly to storage.
- Location ingestion, reporting, work lifecycle, notices, map queries and safety events are separate service modules.
- Real-time updates should use WebSocket/SSE/polling abstraction so transport can change without rewriting UI.
- Offline mobile actions use an idempotency key so reconnect does not create duplicates.

12. Required screens and routes

Desktop

Route / view	Purpose
/	Tom map / manager map default.
/work	Operational queue/list + map context.
/safety	Active lone-worker sessions, alerts, history and configuration.
/notices	Publish/manage expiring notices.
/registry	Zones, anchors, assets and QR management.
/settings	Users, workspaces, permissions, alert rules, integrations/config.

Mobile/PWA

Route / state	Purpose
/field	Worker home: current location, status, nearby, Report Here, Help.
/field/report	GPS-first report flow.
/field/lone-work	Start/active/end lone-work session.

/a/:anchorId	QR anchor/asset landing; active notice first.
/a/:anchorId/report	Anchor-context report.
/report/:id/sent	Confirmation / queued state.

13. Technical architecture

Recommended web stack follows the previous build direction: Next.js App Router + React + TypeScript + Tailwind/design tokens + PWA/offline support. Backend choice may vary, but the following capabilities are required.

Capability	Requirement
Auth	Role-based manager/owner authentication. Field report path can remain login-free; worker presence/lone work requires known identity/device association.
Map	Satellite/hybrid provider with clustering, custom layers, zone geometry and map selection events.
Geolocation	Permission-aware location service with accuracy, timestamp, source and stale-state logic.
Offline	Service worker caching + IndexedDB outbox for reports/check-ins that can safely queue.
Media	Client compression; durable object storage; signed upload flow preferred.
Realtime	Push map/state changes to management view; degrade to polling.
Notifications	Provider abstraction for email/SMS/push; log send and acknowledgement state.
Spatial queries	Point-in-zone, nearest anchor/asset, distance, bbox/viewport querying; geospatial DB extension recommended.
Audit	Append-only event history for lifecycle and safety transitions.

14. Phased build plan

Build order Foundation → spatial registry → mobile reporting → Tom map → work loop → lone-worker safety → intelligence/polish. Do not attempt every feature at once.

Phase 0 — Foundation and product skeleton

16. Create repository, environments, TypeScript strict mode, linting/testing and CI.
17. Implement design tokens, typography, components, responsive shell and light/dark themes.
18. Define database entities and migrations for Estate/Workspace/Zone/Anchor/Asset/Person/Observation/WorkItem/Notice/LoneWorkSession/SafetyEvent.
19. Seed Raynham demo data and zones sufficient to reproduce both mock-ups.
20. Implement API/service layer and feature flags. No UI component talks directly to DB/local storage except through service abstractions.

Exit criteria: token system renders consistently; seeded registry is queryable; application shells run at mobile and desktop sizes; zero critical accessibility/type errors.

Phase 1 — Spatial registry + mobile reporting MVP

21. Implement zones, anchors/assets and QR URL resolution.
22. Build mobile home with GPS context, working state and nearby items.
23. Build GPS-first Report Here flow and anchor-based variant.
24. Implement camera/file fallback, categories, optional note, auto-location, zone resolution and nearest-object context.
25. Implement IndexedDB offline outbox and idempotent sync.
26. Implement manager receiving/triage of observations and map pins.

Exit criteria: a worker in a field can submit an identified or anonymous report online/offline; Tom sees it at the correct spatial location; a QR scan identifies the exact anchor/asset.

Phase 2 — Tom's spatial operating picture

27. Build whole-estate satellite map and workspace lenses.
28. Implement layers, clustering and zoom-dependent density.
29. Add Estate Status / ALL CLEAR calculation.
30. Add Needs Attention queue linked to fly-to/open-inspector actions.
31. Build inspectors for place/asset/person/observation/work item.
32. Implement time filters and basic spatial history queries.
33. Add global search across people, places, assets and work.

Exit criteria: Tom can open one screen, identify active people/issues/lone-work placeholder states, select any object and understand current context and history without leaving the map.

Phase 3 — Work management + notices

34. Separate Observation from WorkItem and implement triage/create-work flow.
35. Assignment to staff/contractor; status lifecycle; resolution evidence.
36. Contractor link experience for job details and resolution without full account if approved.
37. Notices scoped to estate/workspace/zone/anchor/asset with mandatory expiry.
38. QR scan surfaces active notice before normal actions.
39. Repeat-location/duplicate heuristics produce manager suggestions, never silent destructive merges.

Exit criteria: report → triage → assignment → resolution → retained place/asset history works end to end.

Phase 4 — Lone-worker safety

40. Implement worker identity/device binding and explicit Working/Off Duty states.
41. Prototype background location on the actual estate target devices and document limitations.
42. Implement Lone Work start, activity, expected finish, check-in cadence and end.
43. Implement stale-location handling and explicit last-seen display.
44. Implement Attention → Emergency escalation engine and on-call notifications.
45. Implement Tom/manager lone-worker map details and acknowledgement/resolution workflow.
46. Run failure simulations: no signal, permission revoked, battery saver, browser killed, device left stationary, missed check-in.

Exit criteria: safety workflow passes estate-agreed operational tests. If background reliability fails required thresholds, pause production safety claims and move beacon capability to a native wrapper/app track.

Phase 5 — Intelligence, history and hardening

47. Generate Estate Brief from deterministic operational data first; add LLM summarisation only behind factual source references.
48. Add recurring hotspot insights by zone/place/asset/category and 7/30-day time filters.
49. Add spatial command/search enhancements where useful.
50. Performance optimisation, observability, backups, privacy retention, accessibility and security review.

51. Pilot with a small Raynham cohort, collect false-alert/report-completion/location-quality metrics, then iterate before estate-wide rollout.

15. AI development-agent execution rules

An AI coding agent should treat this section as operational instructions.

52. Build only the current phase. Do not pre-emptively add “enterprise” features.
53. Before coding a screen, identify the user, job-to-be-done, spatial objects, states, permissions and offline behaviour.
54. Use the approved mock-ups for visual hierarchy; use this PRD for behaviour when they differ.
55. Every new feature must attach to at least one of: Place, Person, Thing, Work, Risk.
56. Every location value must include timestamp/source/accuracy or fallback metadata where applicable.
57. Never use red outside true emergency entry/state.
58. Never block reporting/help behind signup. Identity may improve attribution; lone-work requires known identity.
59. Write tests for happy path + permission denied + no signal + stale data + duplicate submission + role denial.
60. Use feature flags for incomplete safety/notification/external-service capabilities.
61. At phase end, run acceptance tests and produce a short implementation note listing assumptions, TODOs and known platform constraints before continuing.

16. Acceptance criteria — release gate

Mobile

- At 360px and 390px, primary actions are readable, one-handed and >=44px targets.
- Worker location always shows update age; stale/unavailable states are distinct.
- Report can be completed without typing and without account login.
- Location permission denial does not dead-end reporting; QR context/explicit map pin/fallback is used.
- Offline report confirms locally, queues once and syncs idempotently.
- QR notice appears before normal actions when active.
- Lone-work session requires identified person and shows exact next check-in/expected finish.
- Help flow records last-known location and clearly states what notification action occurs.

Desktop

- Map remains primary surface at desktop widths; overlays do not obscure the majority of estate context.
- Multiple layers can coexist; clustering prevents pin soup.
- ALL CLEAR calculation is deterministic and documented.
- Needs Attention items navigate to and select spatial context.
- Inspectors show Current / History / Actions and location freshness where relevant.
- Workspace lens changes visible data without creating separate siloed systems.
- Time range changes map/history results predictably.

Quality, safety and accessibility

- WCAG 2.2 AA target; status never conveyed by colour alone.
- No raw design palette values inside feature components where semantic tokens exist.
- No production claim of certified fall detection or guaranteed continuous GPS without validated capability.
- No off-duty location exposed to Tom/managers.
- Safety transitions, acknowledgements and notification attempts are auditable.
- Critical API writes are authenticated/authorised or use narrowly scoped signed public tokens where login-free flow requires it.

17. Decisions still required before production

Decision	Owner / reason

Exact lone-worker SOP and escalation timings	Raynham operations / safety lead. Product cannot invent emergency policy.
Location update frequency and retention	Tom + staff policy/privacy advice; balances safety, privacy and battery.
PWA vs native wrapper for background beacon reliability	Engineering after real-device Phase 4 validation.
Notification channels and on-call rota	Estate operations.
What3Words service/API implementation and fallback	Engineering/product; retain GPS as canonical coordinate.
Map provider/licensing and estate boundary data	Product/engineering.
Contractor identity and resolution security model	Product/engineering.

18. Build seed — canonical demo scenario

Use this scenario for development and demos so all teams build against the same story:

62. David Reeves starts Working in North Field; phone reports a recent location.
63. David sees Gate 4 (62m), Water Trough (110m), and an existing fence-damage issue (150m).
64. David finds new fence damage, taps Report Something Here, adds photo + Fence category and submits.
65. Tom's map immediately shows the new fixed incident in North Field while David's person marker remains independently mobile.
66. The issue enters Needs Attention and manager creates/assigns a Work Item.
67. David later starts Lone Working in North Woods, expected finish 15:30, next check-in 14:45.
68. Tom sees David in amber Lone Working state with last update age. David checks in via I'm OK.
69. Work is resolved with evidence; Gate/North Field history retains the event. Estate Brief can later surface recurrence if similar reports accumulate.

19. Definition of done

The platform is successful when A worker can interact with Raynham from where they stand, and Tom can understand Raynham from above — with location, work and safety joined into one calm spatial operating picture.

This PRD intentionally keeps the product narrow. Anything that does not improve Place, Person, Thing, Work or Risk in the spatial operating picture should be challenged before it is added.

Appendix A — Source basis and supersession

This v2.0 build specification updates the July 2026 North Star PRD and mobile personnel reporting flow. It preserves their strongest rules: phone-based lone-worker replacement, QR anchors, login-free reporting, offline-first field ergonomics, red-for-emergency-only, workspace lenses and the map as the default management surface. It strengthens the model by making the phone a work-time spatial beacon, separating observation from work, introducing explicit location

freshness/source, composable layers, place/asset history, map-first Tom experience, and a phased technical validation path for lone-worker beacon reliability.